

SMART INDIA HACKATHON 2026

TATPAR

तत्पर • “ever-ready”

Readiness Assurance Platform for Air Fleets

Others predict failures. TATPAR assures readiness.



AI prognostics • sustainment digital twin • fleet optimisation • offline & sovereign

PROBLEM STATEMENT ID

26249

PROBLEM STATEMENT TITLE

Air Power – Predictive Maintenance & Fleet Availability

ORGANISATION

Ministry of Defence • Defence Services Staff College

THEME

Transportation & Logistics

PS CATEGORY

Software

TEAM ID

To be filled

TEAM NAME

To be filled

Proposed Solution

TATPAR joins health monitoring, technical records, spares and repair-agency data, then plans flying, maintenance and spares so that the required number of aircraft is mission-capable on the day of need — with stated confidence.



“How many aircraft on day D — and how sure?”

Readiness Forecast

A Fleet Twin of aircraft, spares, BRD/HAL repair pipeline, bays and manpower runs 200 Monte-Carlo futures → P10–P90 aircraft per day and P(meet task).



“What must I do today to have N ready?”

Readiness-Backward Planner

Optimises flying hours per tail, check timing, spares transfers, depot priority and cannibalisation; shows each action’s readiness gain; a human approves.



“Where is readiness lost — and why?”

Readiness-Loss Waterfall

Aircraft-days lost to spares, scheduled and unscheduled work, depot and manpower, with the days each lever could recover.

What is unique

Twin of the sustainment system — not a 3D model of one aircraft

Plans backwards from the requirement — each action scored by P(meet), checked against hidden truth

Live data fabric — HUMS, tech log, IMMOLS, BRD/HAL imports change the plan

Calibrated RUL — conformal 90 % intervals + engine-module attribution

Multi-echelon spares (VARI-METRIC); NFF, rogue-unit & cannibalisation analytics

Offline, federated, signed ledger; Indian base dust from real CAMS data

PS ASKS → TATPAR

AI/ML predictive maintenance
calibrated engine RUL, LRU survival, NFF & rogue units

IoT / aircraft health monitoring
HUMS downloads imported live via an edge gateway

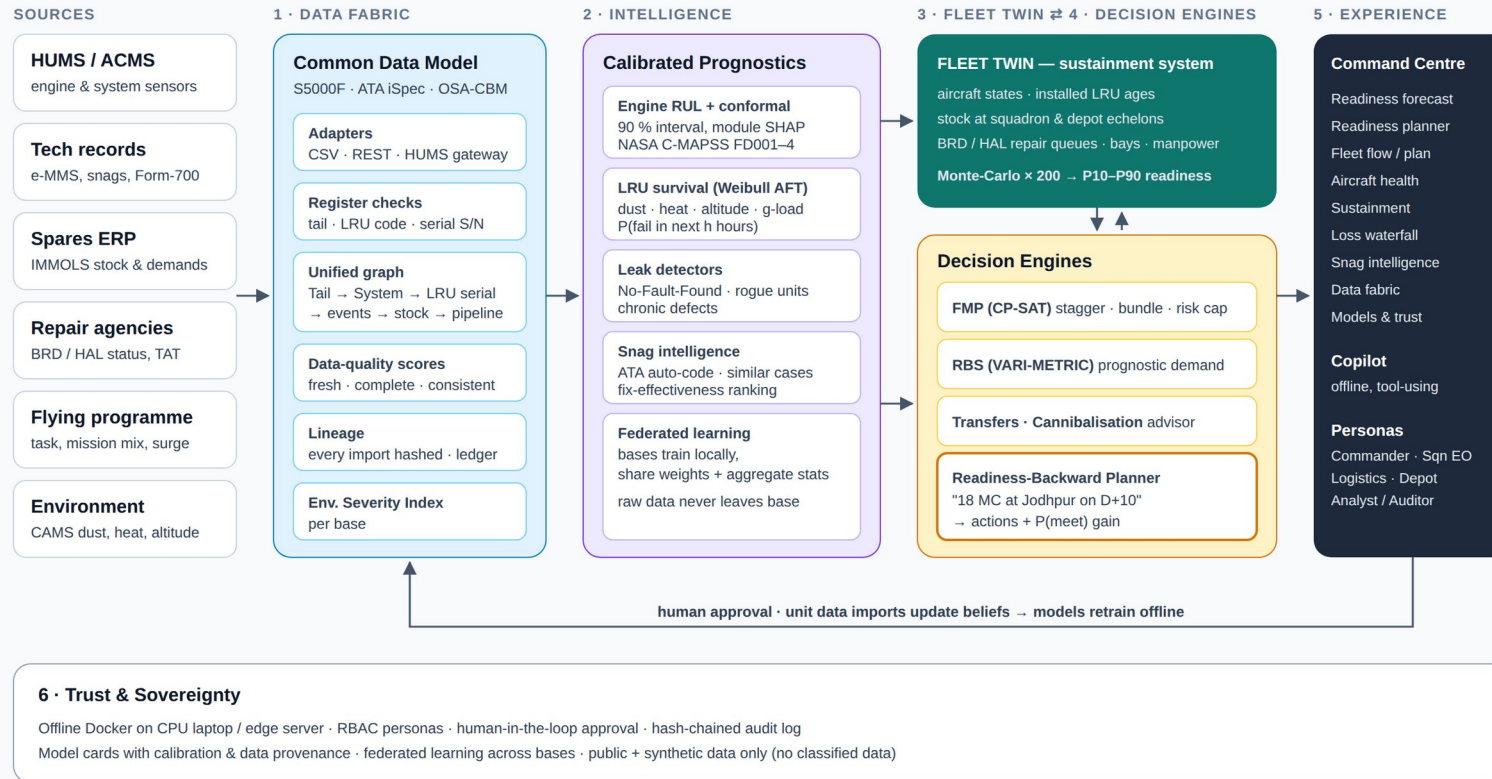
Digital twin
Fleet Twin of aircraft, spares, BRD/HAL, bays, crews

Integrated analytics platform
4 data sources joined; 8-board ops room; signed ledger

Technical Approach

TATPAR — Readiness Assurance Architecture

Integrate → Predict (calibrated) → Simulate (Fleet Twin) → Optimise → Assure → Learn · runs fully offline



Technology stack

Prognostics

LightGBM quantile + conformal (CQR), SHAP, Weibull AFT (lifelines)

Twin & optimisation

NumPy Monte-Carlo Fleet Twin, OR-Tools CP-SAT, VARI-METRIC

Data in

CSV contracts + REST for HUMS, tech log, IMMOLS, BRD/HAL; HUMS edge gateway

Platform

FastAPI · React + TypeScript · ECharts · Parquet (Postgres/TimescaleDB in production)

Data

NASA C-MAPSS, MaintNet logbooks, CAMS dust, notional synthetic fleet

Deploy

Docker, fully offline on CPU laptop / edge server

Integrate

Predict

Simulate

Optimise

Approve

Learn

Feasibility & Viability

+20 pts

mission-capable in the prototype's Fleet Twin:
57.7 → 77.9 %, same fleet & spares budget; +16
to +26 under every changed assumption
(notional)

90 %

calibrated engine-RUL interval coverage on NASA
C-MAPSS test sets (82.7 % before conformal
calibration)

0

cloud dependencies — runs air-gapped on a CPU
laptop or edge server; public + synthetic data
only

Challenge / risk	Mitigation built into TATPAR
No access to real IAF data	Public data + synthetic fleet with hidden ground truth; data contracts let a unit import its own HUMS, e-MMS, IMMOLS, BRD exports
Trust in AI recommendations	Calibrated intervals; planner checked against hidden truth; human approval; signed ledger
Classified, air-gapped networks	Fully offline (Docker); federated learning shares only weights + aggregate statistics
Messy legacy records	Row-by-row validation with reasons; data-quality scores; every import hashed and logged
Optimisation run-time	CP-SAT plans a squadron in 0.3–3.5 s, same plan on every machine

0–3 months • Pilot

One squadron's historical HUMS, e-MMS
and IMMOLS exports mapped onto the
contracts; models retrained;
recommendations in shadow mode

3–9 months • One station

Daily signal, planner and orders in use;
BRD/HAL status feed; MC rate measured
against the station's own baseline

9–18 months • Command

All bases; federated learning over
AFNET; PKI-signed ledger; integration
with IMMOLS / e-MMS

Cost (indicative): one CPU edge server per station (~₹3–5 lakh); open-source stack, no licence fees. **Built and tested:** 8-board ops room, 50 backend tests + 24 browser checks in CI • github.com/DeepakSinghhh/air-power

Impact & Benefits

~40 %

of fighters reported unserviceable at any time
(post-Op Sindoor commentary, Swarajya 2025)

20–50 %

of avionics removals end as No-Fault-Found —
wasted spares and repair slots (Raza 2018)

≈ 13

more aircraft mission-capable every day from a
64-aircraft fleet in the prototype twin — same
spares budget (notional data)

Who benefits



Commander

N aircraft on day D, with a stated confidence



Squadron Engineering Officer

Which tails fly, which go in, what to bundle



Logistics Officer

What to stock, move or expedite — and where



BRD / Depot

Which repairs add the most readiness first



Operational

More aircraft from the same fleet; surge plans with known odds



Economic

≈13 aircraft of availability ≈ ₹14,600 cr of new airframes at the 2024 HAL price (illustrative); spares bought for availability per rupee



Strategic

Sovereign, offline software; supply-risk visibility for indigenisation



Safety & people

Fewer cannibalisations; chronic defects caught early

Research & References

IAF context

- Business Standard (2014) — Su-30MKI serviceability 48–55 %
- CAPS Issue Brief 08/25 — AI in IAF logistics, IMMOLS, e-MMS
- The Week (Aug 2026) — ML for Sukhoi and Tejas fleets, edge AI
- Defense Mirror — 11 BRD Ojhar Su-30MKI overhaul
- Swarajya (2025) — Op Sindoor: ~40 % unserviceable
- MoD–HAL contract, 12 Su-30MKI, ₹13,500 cr (Dec 2024)

Operations research

- Kozanidis — Flight & maintenance planning, Hellenic AF
- Peschiera et al. — Long-term FMP, French AF (arXiv 2001.09856)
- Mattila & Virtanen (2014) — Fighter maintenance sim-optimisation
- Sherbrooke — VARI-METRIC, Operations Research (1986)
- GAO-02-86 — Military aircraft cannibalisations

Prognostics & AI

- Saxena et al. (2008) — NASA C-MAPSS turbofan dataset
- LSTM quantile + conformal RUL — Sensors 26(7) 2249 (2026)
- Federated RUL prognostics — arXiv 2506.00499
- MaintNet aviation logbooks — COLING 2020
- DST-Group-TR-3367 — dust-driven engine degradation
- Raza, Aerospace 5(2):38 (2018) — avionics NFF 20–50 %

Gap analysis: 7 public PS-26249 projects reviewed (NIRANTAR, AERO-READY, AeroTwin-AI, AirPower, SIH_249, PREDIX, VAYU SEWA) — none model flight & maintenance planning, multi-echelon sparing or probabilistic readiness. Prototype figures use a notional fleet; no classified data.